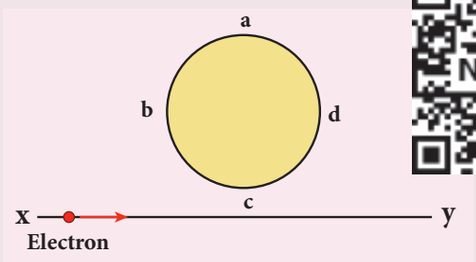
I Multiple Choice Questions

1. An electron moves on a straight line path XY as shown in the figure. The coil *abcd* is adjacent to the path of the electron. What will be the direction of current, if any, induced in the coil?

(NEET 2015)



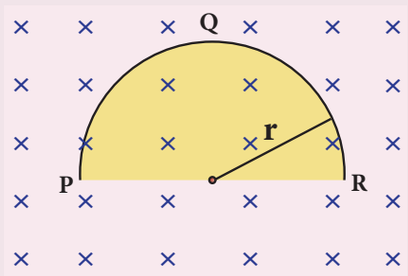
(a) The current will reverse its direction as the electron goes past the coil

(b) No current will be induced

(c) abcd

(d) adcb

2. A thin semi-circular conducting ring (PQR) of radius r is falling with its plane vertical in a horizontal magnetic field B, as shown in the figure.



The potential difference developed across the ring when its speed υ, is

(NEET 2014)

(a) Zero

(b) $\frac{B\upsilon\pi r^2}{2}$ and P is at higher potential

(c) $\pi rB\upsilon$ and R is at higher potential

(d) $2rB\upsilon$ and R is at higher potential

3. The flux linked with a coil at any instant t is given by $\Phi_B = 10t^2 - 50t + 250$. The induced emf at t = 3 s is

(a) −190 V (b) −10 V

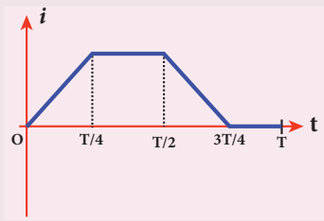
(c) 10 V (d) 190 V

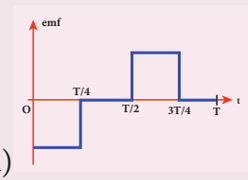
4. When the current changes from +2A to −2A in 0.05 s, an emf of 8 V is induced in a coil. The co-efficient of self-induction of the coil is

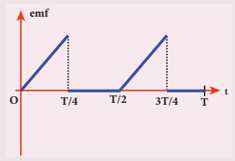
(a) 0.2 H (b) 0.4 H

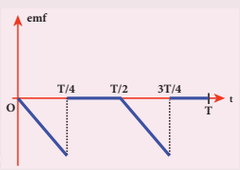
(c) 0.8 H (d) 0.1 H

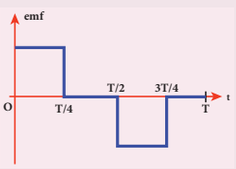
5. The current *i* flowing in a coil varies with time as shown in the figure. The variation of induced emf with time would be (NEET 2011)



(a)


(b)


(c)


(d)


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